

1. Variables & Data Types

Type	Example	Notes
Integer	x = 10	Whole numbers
Float	y = 3.14	Decimal numbers
String	s = 'Hello'	Text
Boolean	b = True	True or False
List	lst = [1,2,3]	Ordered, mutable
Tuple	tup = (1,2,3)	Ordered, immutable
Dictionary	d = {'a':1}	Key-value pairs
Set	st = {1,2,3}	Unordered, unique elements
NoneType	n = None	Represents no value

```
class Person:
def __init__(self,name,age):...
def greet(self): ...
```

8. Modules & File I/O

```
import math
math.sqrt(16)
with open('file.txt','w') as f: f.write('Hello')
with open('file.txt','r') as f: f.read()
```

2. Strings

```
s = 'Python'
# Access & slicing
s[0], s[1:4]
# Methods
s.upper(), s.replace('P','J')
```

3. Lists & Tuples

```
lst = [1,2,3]
lst.append(4), lst.remove(2)
tup = (1,2,3)
```

4. Dictionaries & Sets

```
d = {'a':1, 'b':2}, d['c'] = 3
s = {1,2,3}, s.add(4), s.remove(2)
```

5. Conditionals & Loops

```
x = 10
if x>5: ... elif x==5: ... else: ...
for i in range(5): ...
while x>0: x-=1
```

6. Functions

```
def add(a,b): return a+b
square = lambda x: x**2
```

7. Classes & OOP